Detecting Multilingual & Multicultural & Multi Event Polarization of Social Media

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Introduction

In this era where the vast majority of human interactions take place through social media, mass online polarization is becoming an increasingly serious problem. Polarization, which can be described under headings such as racism, religious manipulation, politics, and sexism, has until recently been attempted to be regulated by only human intervention. Current evidence indicates that achieving 80% moderation coverage requires human review
of at least 10% of all daily content which will be translated to 5.1 million tweets per day for English alone. At more practical review rates of 1%, only 20-40% of polarizing content is successfully moderated, leaving the majority of harmful discourse unaddressed [1]. Utilizing developing artificial intelligence technology effectively in this regard has become essential in terms of time and resource management.

Methods

Our goal was to find the most accurate one among those with multilingual models, and thus seek a solution to the polarization problem.

Training Candidates:

- mBERT (bert-base-multilingual-cased)
- DistilmBERT (distilbert-base-multilingual-cased)
- XLM-RoBERTa (XLM-R)
- mDeBERTa
- RemBERT

Evaluation:

1. Diagnostics Metrics:
   a. Accuracy: Vibe Check
   b. Precision: Correctness
   c. Recall: Completeness
   d. Per-Class F1: Class specific correctness
   e. Macro F1: Mean of Class F1
2. Quantitative Metrics:
   a. LIME (Local Interpretable Model-agnostic Explanations)
   b. SHAP (SHapley Additive exPlanations)

The models trained with the dataset, fine tuned, and then using proper metrics, a discussion studied.

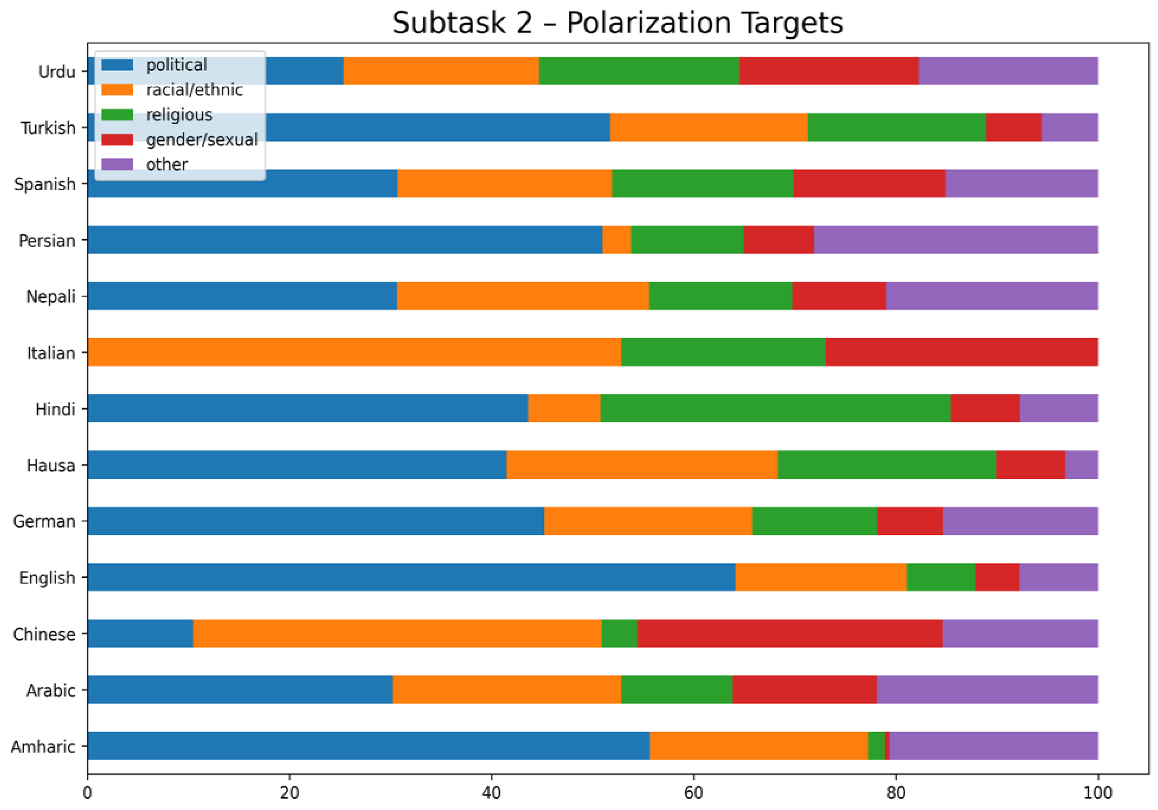
Dataset & Information

The dataset used in this project comes from the POLAR @ SemEval-2026 Task 9 [2]. The dataset includes online platforms such as news websites, Reddit, blogs, Bluesky, and regional forums. It is divided to 3 subtask forms:

- Polarization Detection: Binary classification (Polarized vs. Non-polarized)
- Polarization Type Classification: Political, Racial/Ethnic, Religious, Gender/Sexual, Other
- Manifestation Identification: Stereotype, Vilification, Dehumanization, Extreme language, Lack of empathy, Invalidation

Main Problem of the Dataset: Imbalanced Classes

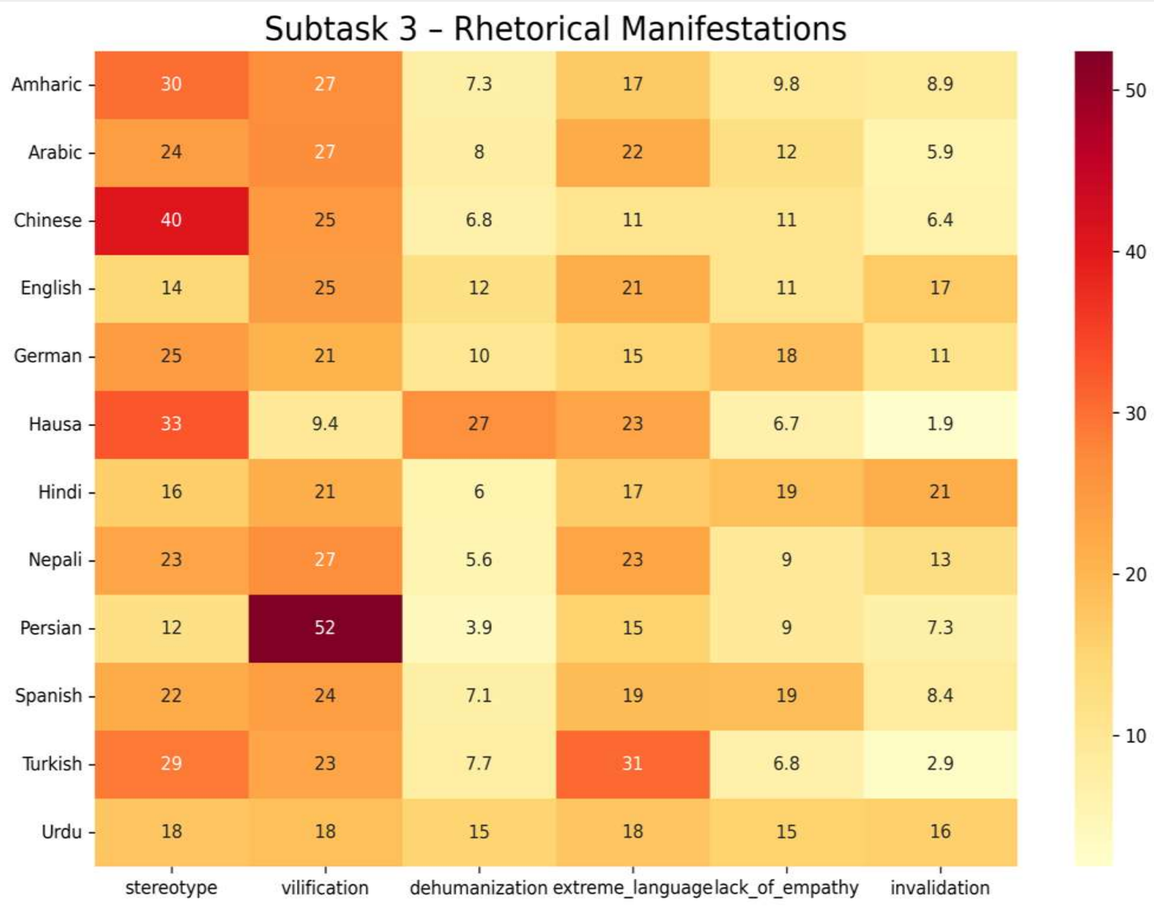
Language	Type	Samples	Polarized %	Avg Length	Vocab
Amharic	Semitic	3,332	75.6% (Imbalanced)	15.4 words	18,989
Arabic	Semitic	3,380	44.7% (Balanced)	16.7 words	21,796
Chinese	Logographic	4,280	49.6% (Balanced)	33.8 chars	3,206
English	Analytic	2,676	37.4% (Imbalanced)	12.4 words	6,867
German	Fusional	3,180	47.5% (Balanced)	17.4 words	10,303
Hausa	Chadic	3,651	10.7% (Imbalanced)	19.7 words	11,975
Hindi	Fusional	2,744	85.5% (Imbalanced)	26.4 words	8,110
Italian	Fusional	3,334	41.0% (Balanced)	23.3 words	13,637
Nepali	Fusional	2,005	50.3% (Balanced)	17.0 words	8,251
Persian	Analytic	3,295	74.1% (Imbalanced)	19.3 words	12,234
Spanish	Fusional	3,305	50.2% (Balanced)	10.8 words	7,560
Turkish	Agglutinative	2,364	48.9% (Balanced)	20.9 words	20,242
Urdu	Fusional	2,849	69.4% (Imbalanced)	18.2 words	9,294



Strongly imbalanced languages:

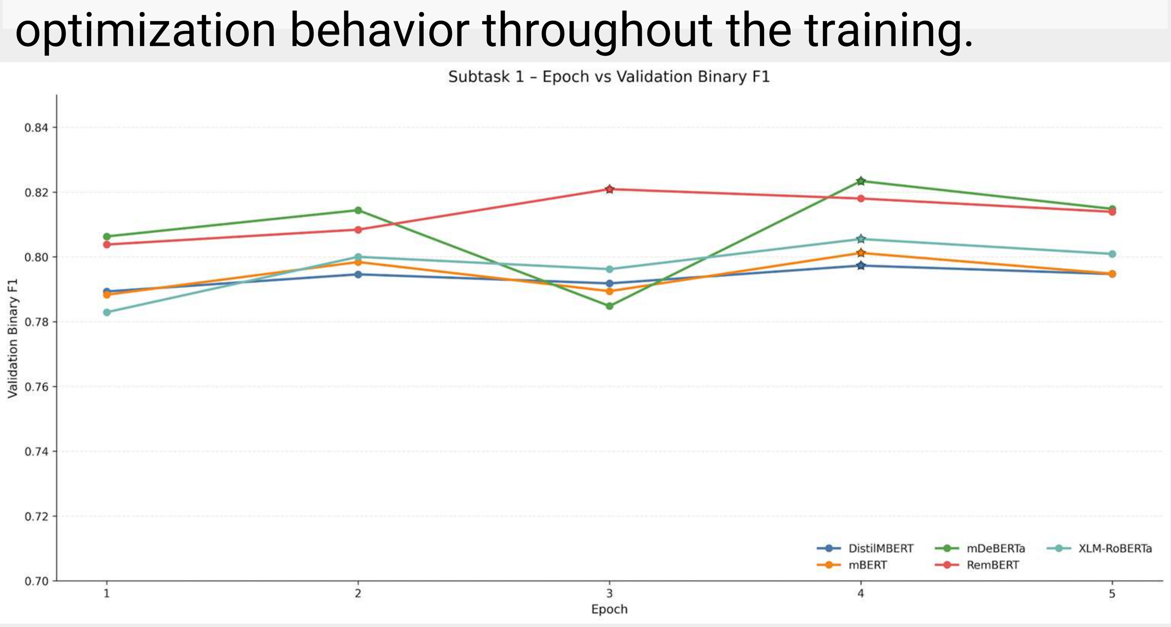
- Italian
- Hindi
- Hausa
- Chinese (Low vocabulary)
- Persian

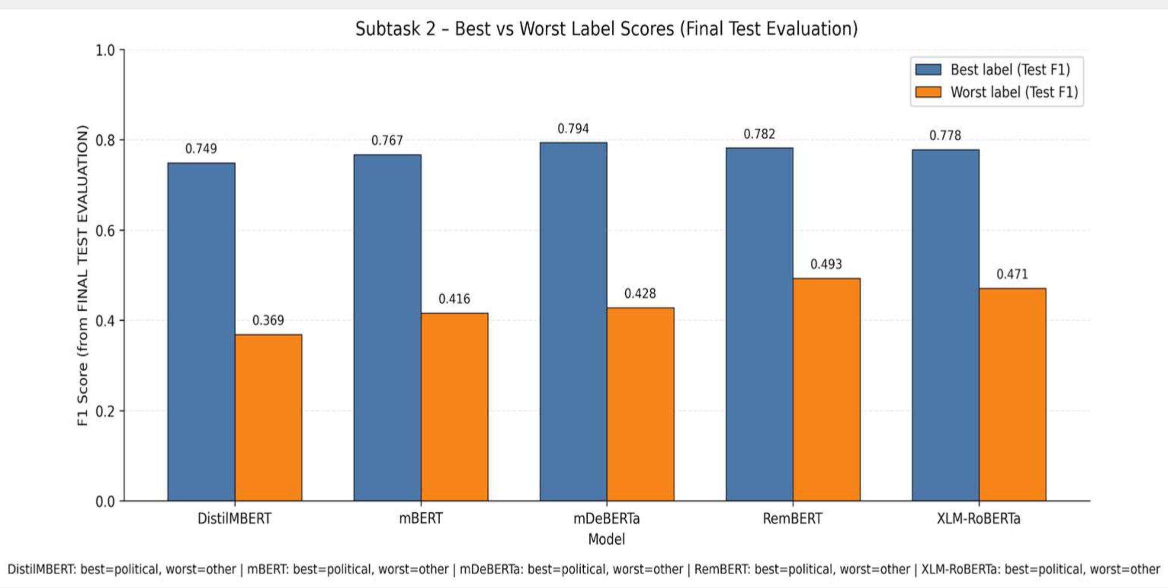
Some languages, like Italian and Hausa, show strong emphasis on specific categories, whereas others display a more balanced distribution across multiple types. This variation highlights that each language reflects different social or cultural issues. [3]



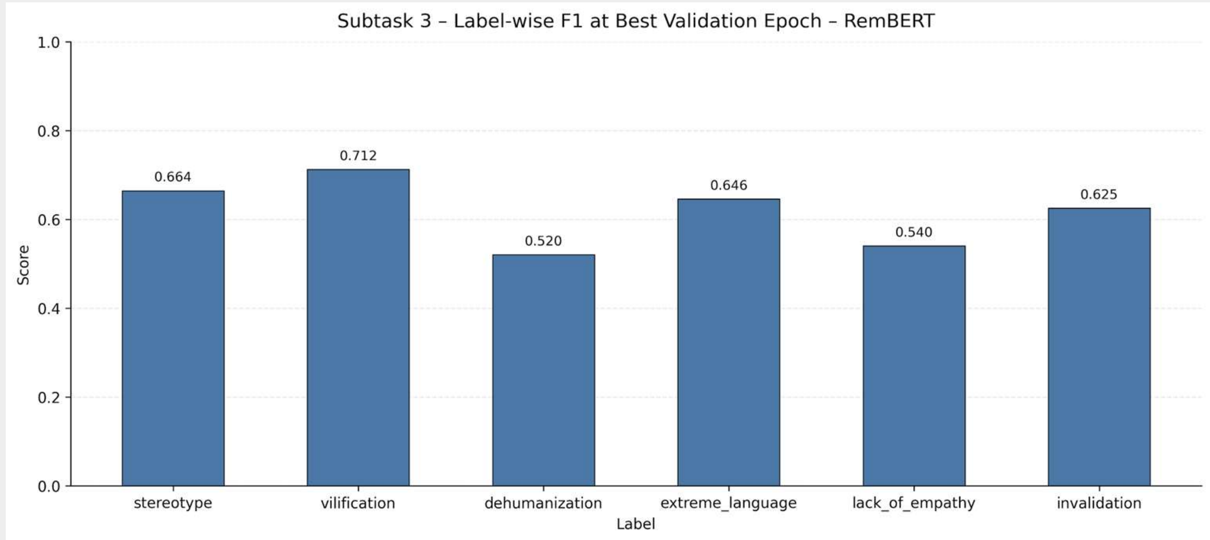
Results

For Subtask 1, all models reach near-peak validation Binary F1 scores within early epochs, with only minor fluctuations thereafter. This reflects stable optimization behavior throughout the training.





In Subtask 2, across all models, *political* polarization achieves the highest F1 scores, while the *other* category performs worst. This reflects the impact of label ambiguity and class imbalance on Macro F1.



Lastly for Subtask 3, *vilification* and *stereotype* achieve the highest F1 scores for RemBERT model, while *dehumanization* and *lack of empathy* remain more challenging. This indicates difficulties in detecting implicit or rare manifestations.

Expected vs Actual Results

Expected Results:

Our primary hypothesis was that large, fine-tuned multilingual transformer models (such as XLM-RoBERTa-Large or RemBERT) on the POLAR dataset will significantly outperform traditional baselines and zero-shot approaches. This is based on the results of the work of Naseem et al [4].

Actual Results:

The experimental results partially confirm this hypothesis. Larger multilingual models such as RemBERT achieve the strongest overall performance across Subtasks 2 and 3. However, for Subtask 1 mDeBERTa slightly performs better than RemBERT. This can be explained by binary classification task in Subtask 1. Otherwise, performance gains over smaller models are limited, especially in fine-grained tasks that suffer from label ambiguity and class imbalance.

Conclusion

In this project, a systematic evaluation of multilingual transformer models is conducted on the POLAR benchmark. Binary polarization detection is demonstrated to be a stable and well-solved task across all evaluated models. In contrast, fine-grained polarization type and manifestation detection remain challenging due to label ambiguity and class imbalance. Although larger models such as RemBERT outperform smaller architectures, model scaling alone does not overcome dataset-level limitations. These findings indicate that dataset design and label formulation are key factors, outweighing the impact of model size.

References

[1] Manuel Tonneau, Diyi Liu, Niyati Malhotra, Scott A. Hale, Samuel P. Fraiberger, Victor Orozco-Olvera, and Paul Röttger. Hateday: Insights from a global hate speech dataset representative of a day on twitter, 2025.

[2] “Subtasks - POLAR @ SemEval-2026,” Github.io, 2026. https://polar-semeval.github.io/tasks.html (accessed Jan. 06, 2026).

[3] “SemEval 2026 Task 9 - Subtask 2: Multilingual Text Classification Challenge - Polarization Type Classification - Codabench,” Codabench.org, 2026. https://www.codabench.org/competitions/10669/ (accessed Jan. 06, 2026).

[4] U. Naseem et al., “POLAR: A Benchmark for Multilingual, Multicultural, and Multi-Event Online Polarization,” arXiv (Cornell University), May 2025, doi: https://doi.org/10.48550/arxiv.2505.20624.